

computer memories.

High-energy particles ionise the medium through which these pass, leaving behind a wake of electron-hole pairs. These pairs can change the state of a memory cell or a logic flip-flop. As a result, a radiation strike might change not just the state of a memory cell but also the design of the circuit it controls, potentially leading to catastrophic failure.

Engineers only have the option to use triple modular redundancy within the subsystems they design using these parts. This increases the cost and development time and still leaves gaps in the test methodology. Space-oriented components can provide greater levels of protection; this not only simplifies the system design but also improves the overall reliability—the key criterion for space agencies and satellite operators to minimise damaging failures.

The electronics aboard Juno is encased in a titanium vault designed to protect components from high levels of radiation. But even with this shielding, the spacecraft is expected to sustain serious damage after a year in Jupiter's orbit. Any loss of control of Juno could leave the spacecraft in the danger of crash.

Advances being made

A five-year project led by the Georgia Institute of Technology has developed a novel approach to space electronics that could change the way space vehicles and instruments are designed. The new capabilities are based on silicon-germanium (SiGe) technology, which can produce electronics that is highly resistant to both wide temperature variations and space radiation.

The team's overall task was to develop a tested infrastructure that included everything needed to design and build extreme-environment electronics for space missions. The result is a robust material that offers important gains in toughness, speed and flexibility. The robustness is crucial for SiGe's ability to function in

Silicon-germanium (SiGe) technology can produce electronics that is highly resistant to both wide temperature variations and space radiation

space without bulky radiation shields or large power-hungry temperature control devices. Compared to conventional approaches, SiGe electronics can provide major reductions in weight, size, complexity, power and cost, as well as increased reliability and adaptability.

The Radiation Hardened Electronics for Space Environments (RHESE) project endeavours to expand the radiation-hardened electronics by developing high-performance devices robust enough to withstand the extreme radiation and temperature levels of the space environment. The project is a part of the Exploration Technology Development Program (ETDP), which funds an entire suite of technologies needed for accomplishing the goals of the vision for space exploration.

NASA's Marshall Space Flight Center (MSFC) manages the RHESE project. RHESE's investment areas include novel materials, design processes to implement radiation hardening, reconfigurable hardware techniques, software development tools, and radiation environment modeling tools.

Near-term emphasis within the multiple RHESE tasks is on hardening field-programmable gate arrays (FPGAs) for use in reconfigurable architectures and developing electronic components using semiconductor processes and materials (such as SiGe) to enhance the tolerance of a device to radiation events and low-temperature environments.

As these technologies mature, the project will shift its focus to developing low-power, high-efficiency total-processor hardening techniques and hardening of volatile and non-volatile memories. This phased approach to distributing emphasis between technology developments

allows RHESE to provide hardened FPGA devices and environmentally-hardened electronic units for mission infusion into early constellation projects.

Once these technologies begin the infusion process, the RHESE project will shift its technology development focus to hardened high-speed processors with associated memory elements and high-density storage for longer-duration missions, such as the Lunar Lander, Lunar Outpost, and eventual Mars exploration missions occurring later in the Constellation schedule.

The individual tasks of RHESE are diverse, yet united in the common endeavour to develop electronics capable of operating within the harsh environment of space. Specifically, the RHESE tasks include SiGe integrated electronics for extreme environments, modeling of radiation effects on electronics, single-event-effects-immune reconfigurable FPGA, radiation-hardened high-performance processors and reconfigurable computing.

Though the tasks are diverse in their specific key performance parameters, these target to accomplish specific goals—improved total ionisation dose tolerance, reduced single-event upset rates, increased threshold for single-event latch-up, increased sustained processor performance, increased processor efficiency, increased speed of dynamic reconfigurability, reduced lower bound of the operating temperature range, increased available levels of redundancy and reconfigurability, and increased reliability and accuracy of radiation effects modeling. ●

The author is in the department of physics at Sant Longowal Institute of Engineering and Technology, Longowal, Punjab